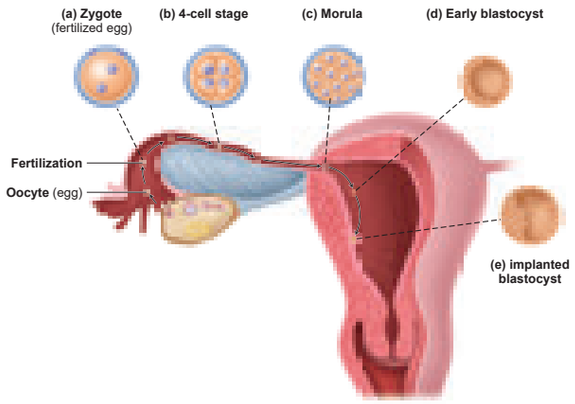


Fig. 2.11 From zygote to blastocyst : passage of growing embryo through the fallopian tube

Twins are two offsprings produced in the same pregnancy.

- Monozygotic (Identical) twins are produced when a single fertilized egg splits into two during the first cleavage. They are of the same sex, look alike and share the same genes.
- Dizygotic (Fraternal) twins are produced when two separate eggs are fertilized by two separate sperms. The twins may be of the same sex or different sex and are non-identical.
- Siamese (United) twins are the conjoined twins who are joined during birth.

2.6 Maintenance of pregnancy and embryonic development

The inner cell mass in the blastula is differentiated into **epiblast** and **hypoblast** immediately after implantation. The **hypoblast** is the embryonic endoderm and the **epiblast** is the ectoderm. The cells remaining in between the epiblast and the endoderm form the mesoderm. Thus the transformation of the blastocyst into a gastrula with the primary germ layers by the movement of the blastomeres is called **gastrulation**. Each germ layer gives rise to specific tissues, organs and organ systems during organogenesis.

The extra embryonic membranes namely the **amnion, yolk sac, allantois** and **chorion** protect the embryo from dessication, mechanical shock and help in the absorption of nutrients and exchange of gases **(Fig. 2.12)**. The amnion is a double layered translucent membrane filled with the amniotic fluid. It provides a buoyant environment to protect the developing embryo from injury, regulates the temperature of the foetus and provides a medium in which the foetus can move. The yolk sac forms a part of the gut and is the source of the earliest blood cells and blood vessels.

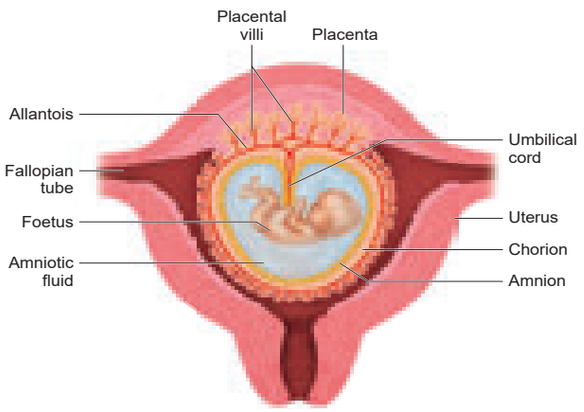


Fig. 2.12 Human foetus within the uterus

The allantois forms a small out pocketing of embryonic tissue at the caudal end of the yolk sac. It is the structural base for the umbilical cord that links the embryo to the placenta and ultimately it becomes part of the urinary bladder. The chorion is the outermost membrane which encloses the embryo and all other membranes and also helps in the formation of the placenta.

The trophoblast cells in the blastocyst send out several finger like projections called **chorionic villi** carrying foetal blood and are surrounded by sinuses that contain maternal blood. The chorionic villi and the uterine tissues form the disc-shaped placenta. **Placenta** is a temporary endocrine organ formed during pregnancy and it connects the foetus to the

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